CASE STUDY 7

SMART PARKING FEE CALCULATOR

Python Programming – Activity-Based Learning

1. Problem Statement

Create a simple Python program to calculate the parking fee based on the type of vehicle and the number of hours it is parked. The program should calculate the fee for a bike, car, or bus and reject invalid inputs.

2. Objectives

• To understand user input in Python.
• To use if-elif-else conditions.
• To calculate a fee using multiplication.
• To display the result clearly.

3. Assumptions

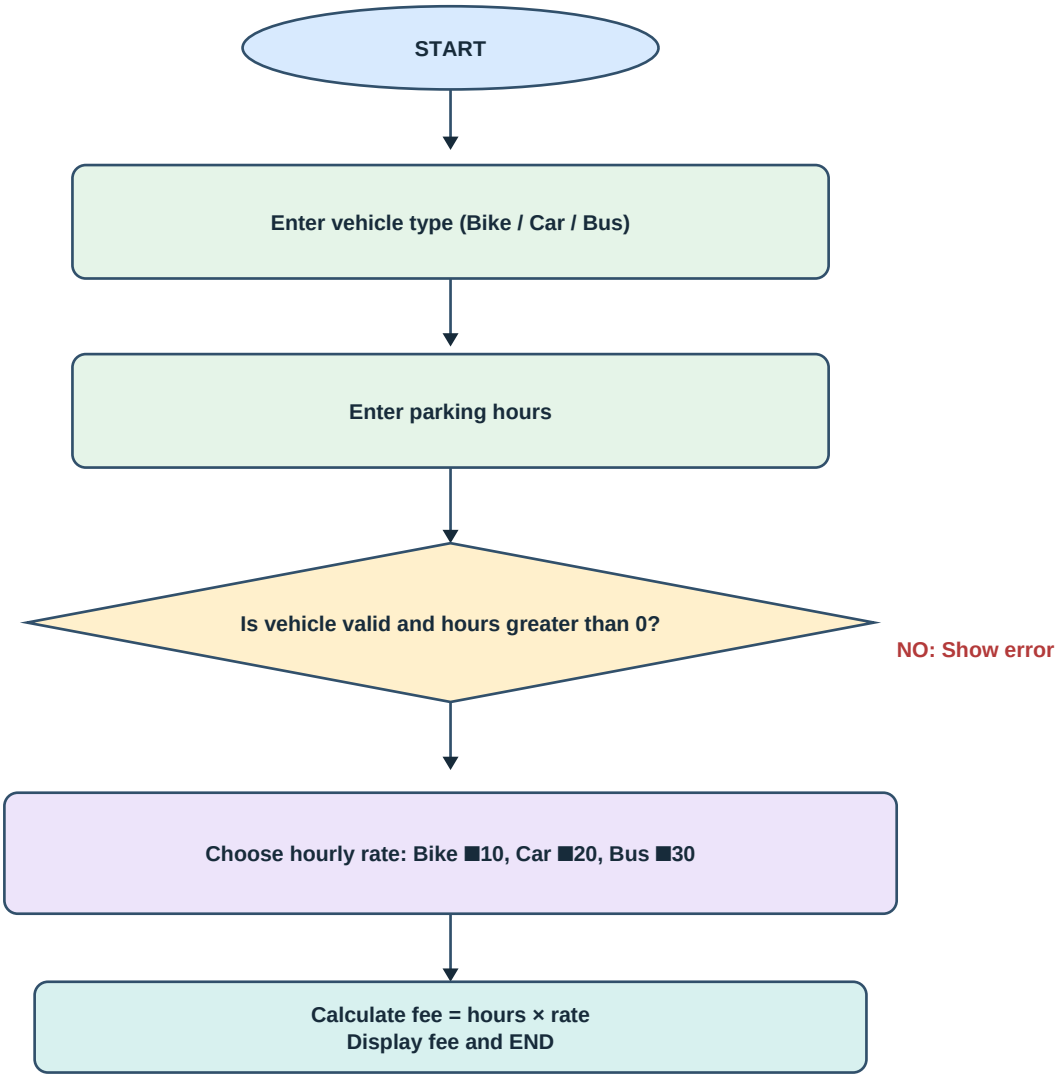
• Vehicle type must be Bike, Car, or Bus.
• Parking hours are entered as a whole number.
• Minimum parking time is 1 hour.
• Bike: ■10/hour; Car: ■20/hour; Bus: ■30/hour.
• No tax or discount is applied.

4. Algorithm

1. Start.
2. Read vehicle type and parking hours.
3. Check that hours are greater than zero.
4. Select the hourly rate based on vehicle type.
5. Calculate fee = hours × rate.
6. Display the parking fee.
7. If input is invalid, display an error message.
8. Stop.

5. Flowchart

Colour flowchart of the program:


START
Enter vehicle type (Bike / Car / Bus)
Enter parking hours
Is vehicle valid and hours greater than 0?
NO: Show error
Choose hourly rate: Bike ■10, Car ■20, Bus ■30
Calculate fee = hours × rate
Display fee and END

6. Pseudocode

```
START
INPUT vehicle
INPUT hours

IF hours <= 0
    PRINT "Invalid parking hours"
ELSE IF vehicle is Bike
    rate = 10
ELSE IF vehicle is Car
    rate = 20
ELSE IF vehicle is Bus
    rate = 30
ELSE
    PRINT "Invalid vehicle type"
    STOP
END IF

fee = hours * rate
PRINT fee
STOP
```

7. Python Source Code

```
vehicle = input("Enter vehicle type (Bike/Car/Bus): ")
hours = int(input("Enter parking hours: "))

if hours <= 0:
    print("Invalid parking hours")
elif vehicle.lower() == "bike":
    fee = hours * 10
    print("Parking fee = Rs.", fee)
elif vehicle.lower() == "car":
    fee = hours * 20
    print("Parking fee = Rs.", fee)
elif vehicle.lower() == "bus":
    fee = hours * 30
    print("Parking fee = Rs.", fee)
else:
    print("Invalid vehicle type")
```

8. Test Cases

Test	Vehicle	Hours	Expected Output
1	Bike	2	Parking fee = Rs. 20
2	Car	3	Parking fee = Rs. 60
3	Bus	2	Parking fee = Rs. 60
4	Car	0	Invalid parking hours
5	Truck	2	Invalid vehicle type

9. Output Screenshots

Sample Run 1 – Bike:

```
Enter vehicle type (Bike/Car/Bus): Bike
Enter parking hours: 2
Parking fee = Rs. 20
```

Sample Run 2 – Car:

```
Enter vehicle type (Bike/Car/Bus): Car
Enter parking hours: 3
Parking fee = Rs. 60
```

Sample Run 3 – Invalid hours:

```
Enter vehicle type (Bike/Car/Bus): Car
Enter parking hours: 0
Invalid parking hours
```

10. Conclusion

This case study helped me understand how to take input, use if-elif-else statements, and perform calculations in Python. The program calculates the parking fee for different vehicles in a simple way.

End of Case Study 7